Development and Accuracy Analysis of Mobile DGNSS using SatRef service

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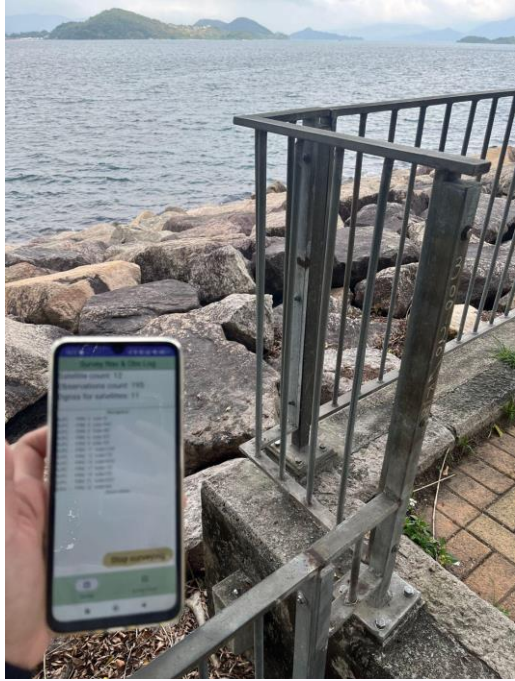

SPP Only	With DGNSS
Easting: 835582.229	Easting: 835582.472
Northing: 821399.032	Northing: 821408.711
Height: 17.529	Height: 5.888
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dN: -9.97	dN: -0.29
dH: +7.47	dH: -4.17

Presentation Outline

1. GNSS background and project motivation
2. DGNSS principles and SatRef service
3. Project objectives
4. Methodology (app workflow and algorithms)
5. Field survey and results
6. Accuracy analysis, discussion, and recommendations

Motivation and Background

- Smartphones increasingly used for positioning
- Problem: Standalone SPP prone to various error sources
- Solution: DGNSS
  - corrects spatially-correlated errors using reference stations
  - simpler implementation
- SatRef provides free, public DGNSS corrections in HK

Project Objectives

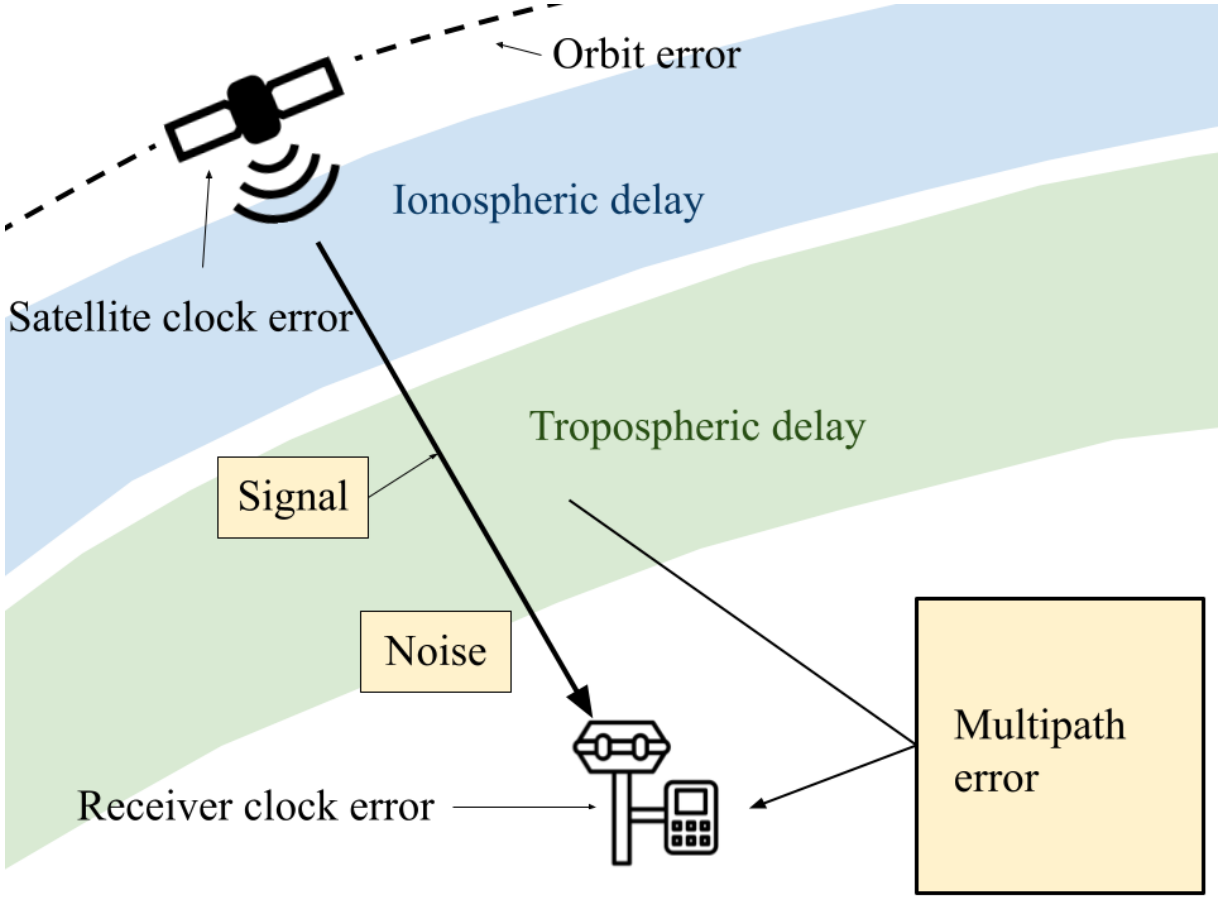
1. Develop a full mobile DGNSS algorithm
2. Evaluate DGNSS accuracy
3. Identify limitations and propose improvements

GNSS Basics

- Global Navigation Satellite Systems
- provide worldwide positioning
- Major constellations: **GPS**, GLONASS, Galileo, BeiDou
- Three segments:
  - Space: satellites
  - Control: control and monitoring stations
  - **User**: receivers

(Kaplan and Hegarty, 2017)

GNSS error sources



(Kaplan and Hegarty, 2017)

Positioning techniques

- **SPP**: simple, meter-level
- PPP: very precise, slow, complex, hardware (carrier-phase)
- RTK: fast and precise, complex, hardware (carrier-phase)
- **DGNSS**: simple, meter-level

-> DGNSS was chosen for this project

-> Compare DGNSS with SPP

(Rosalie et al., 2017; Santerre et al., 2014; El-Rabbany, 2016; Liu et al., 2022)

DGNSS principles

How DGNSS Works: The Correction Chain

- Reference station receives same satellites as rover
- Compares measured pseudorange against theoretical range
- Reference stations sends RTCM containing PRC, RRC, t0
- $P_{corrected} = P_{measured} + PRC + RRC \cdot (t - t0)$

DGNSS Error Cancellation Principle

$$P_{corrected} = \rho + d\rho + c(dt_{recv} - dT_{sat}) + I_{recv} + T_{recv} + M_{recv} + n_{recv} - d\rho - I_{base} - T_{base} - M_{base} - n_{base} \qquad (42)$$

$$P_{corrected} = \rho + c(dt_{recv} - dT_{sat}) + \Delta I + \Delta T + \Delta M + \Delta n$$

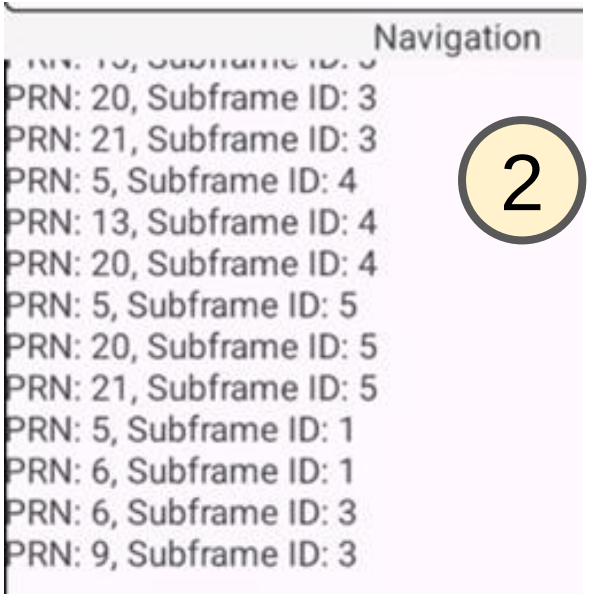
$$P_{corrected} \approx \rho + c(dt_{recv} - dT_{sat}) + \Delta M + \Delta n$$

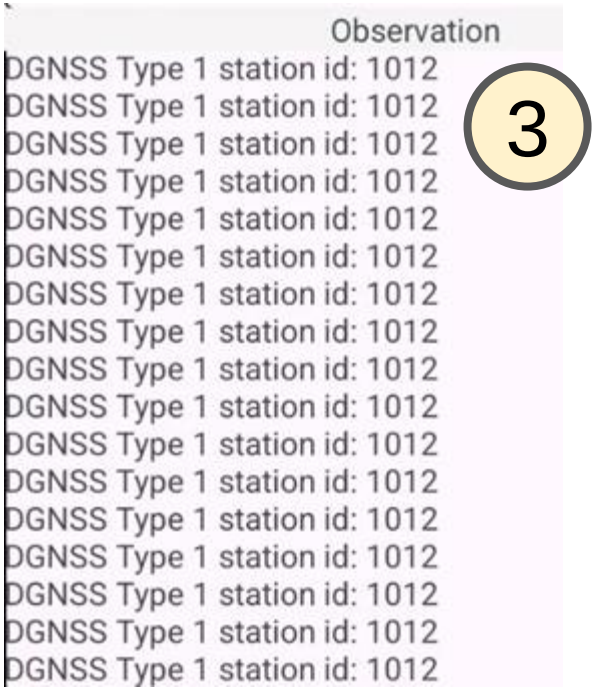
- Cancels orbit error
- Mitigates atmospheric errors (I, T)
- Unknown: Receiver clock error

Methodology

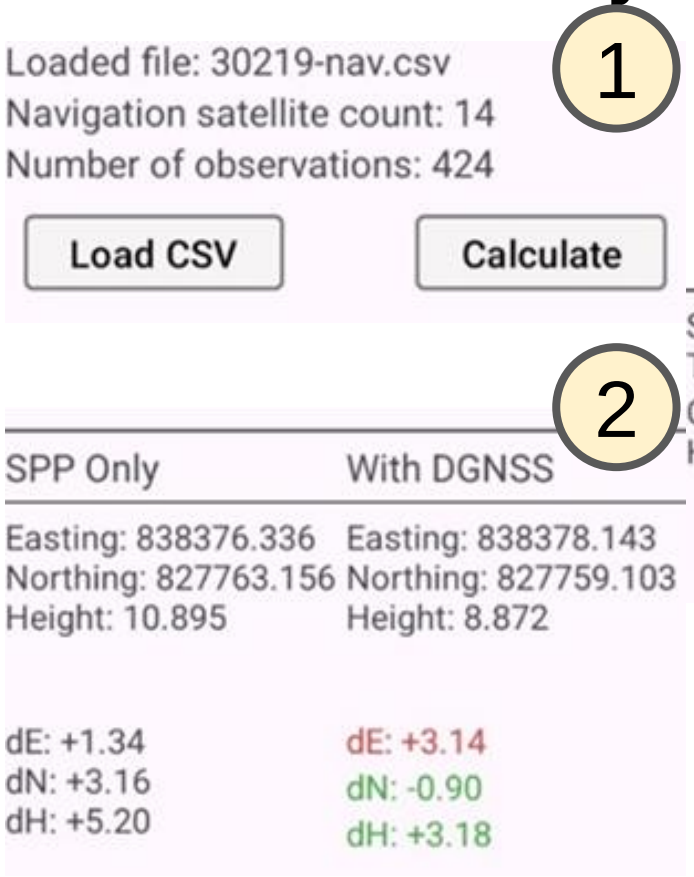
1. Application UI overview
2. Data collection (GNSS and RTCM decoding)
3. SPP corrections (Klobuchar and Saastamoinen models)
4. Weighted least squares and epoch averaging
5. Field survey design

Application UI Overview - Survey Page





Application UI Overview - Survey Store Page



GNSS and RTCM Decoding, A-GNSS

- GNSS subframe parsing
  - adapted from Google
- RTCM v2 Type 1 decoding
  - adapted from RTKLIB, BKG (Federal Agency for Cartography and Geodesy)
- A-GNSS via SUPL
  - .jar library from Android Compatibility Test Suite (CTS)

(Google, 2023; Takasu, 2013; BKG, 2008)

SPP corrections - Klobuchar model

- Klobuchar model on GPS L1
- Ionospheric parameters from subframe 4
- Calculates slant delay from geomagnetic latitude and local time

$$T = 5 * 10^{-9} * F, \quad x \geq \frac{\pi}{2}$$

$$T = (5 * 10^{-9} + \left(1 - \frac{x^2}{2} + \frac{x^4}{24}\right) * amplitude)$$

$$* F, \quad x < \frac{\pi}{2}$$

(Klobuchar, J.A. 1987)

SPP corrections - Saastamoinen model

- ZHD and ZWD using constant meteorology parameters
  - 300K (27°C), 1013 hPa air pressure, 75% humidity
- UNBabc mapping function
  - b, c: constants
  - Accuracy for low elevation satellites (2°)
- Slant delay = mh * ZHD + mw * ZWD

(Saastamoinen, 1972; Davis et al., 1985; Guo and Langley, 2003)

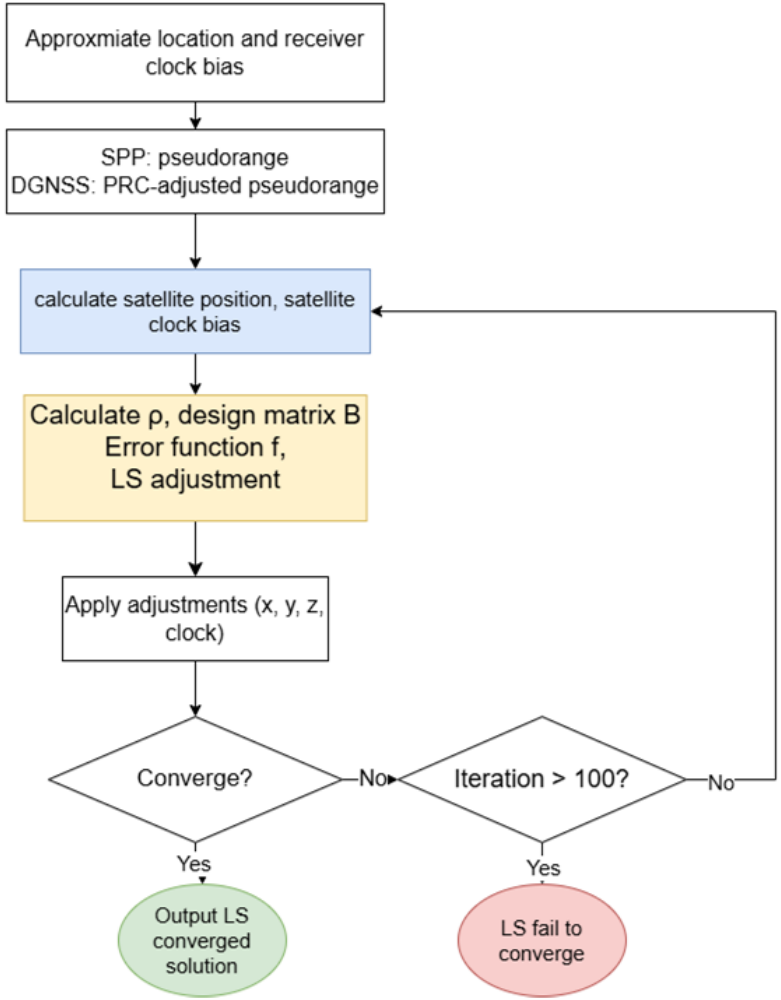
Weighted Least Squares and Epoch Averaging

- Satellite filters: elevation ≥ 15°, SNR > 5 dB
- Weight matrix: Van Dienrendonck (1996) code-tracking jitter model (weight ∝ SNR)
- Iterative LS, converges when $\Sigma|\Delta x,y,z,clock| < 1\times10^{-6}$
- Android clock drift -> solve each epoch independently
- 2-sigma rejection + averaging

(Van Dierendonck, A.J., 1996)

Least squares adjustment flow chart

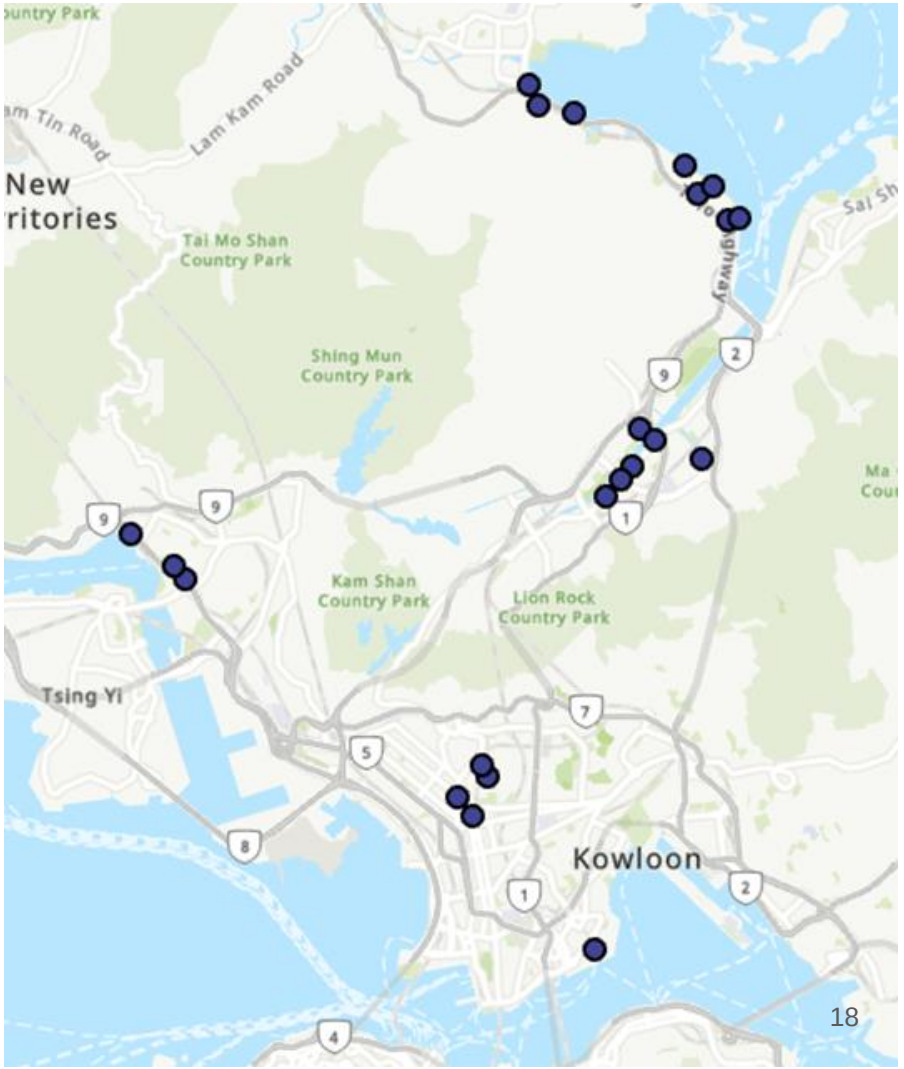
- If **valid** epoch -> add to list for averaging
- If **invalid** epoch -> do not use for averaging
- **all invalid** -> position fail (re-survey)



Field Survey Design

- 22 Hong Kong **geodetic benchmarks** as ground truth
- Different districts, urban and suburbs

- Device: Redmi 14C Android smartphone
- Antenna height: 1m



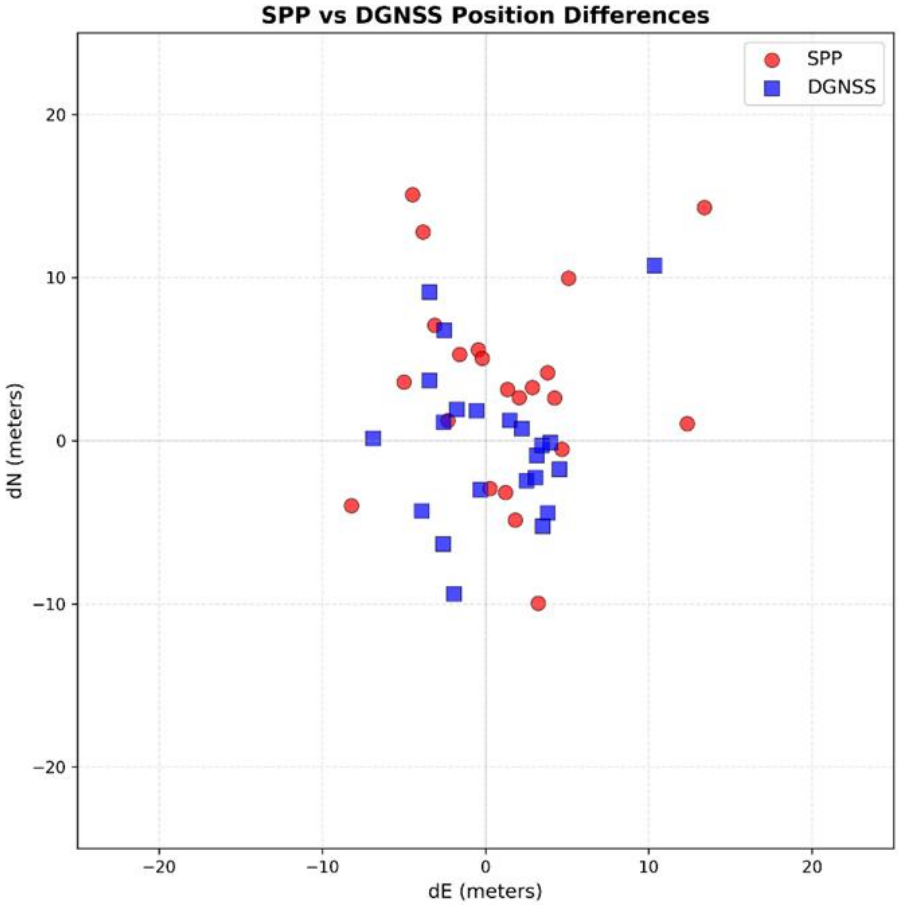
Results

1. Overall accuracy summary
2. Error distribution and scatter plots

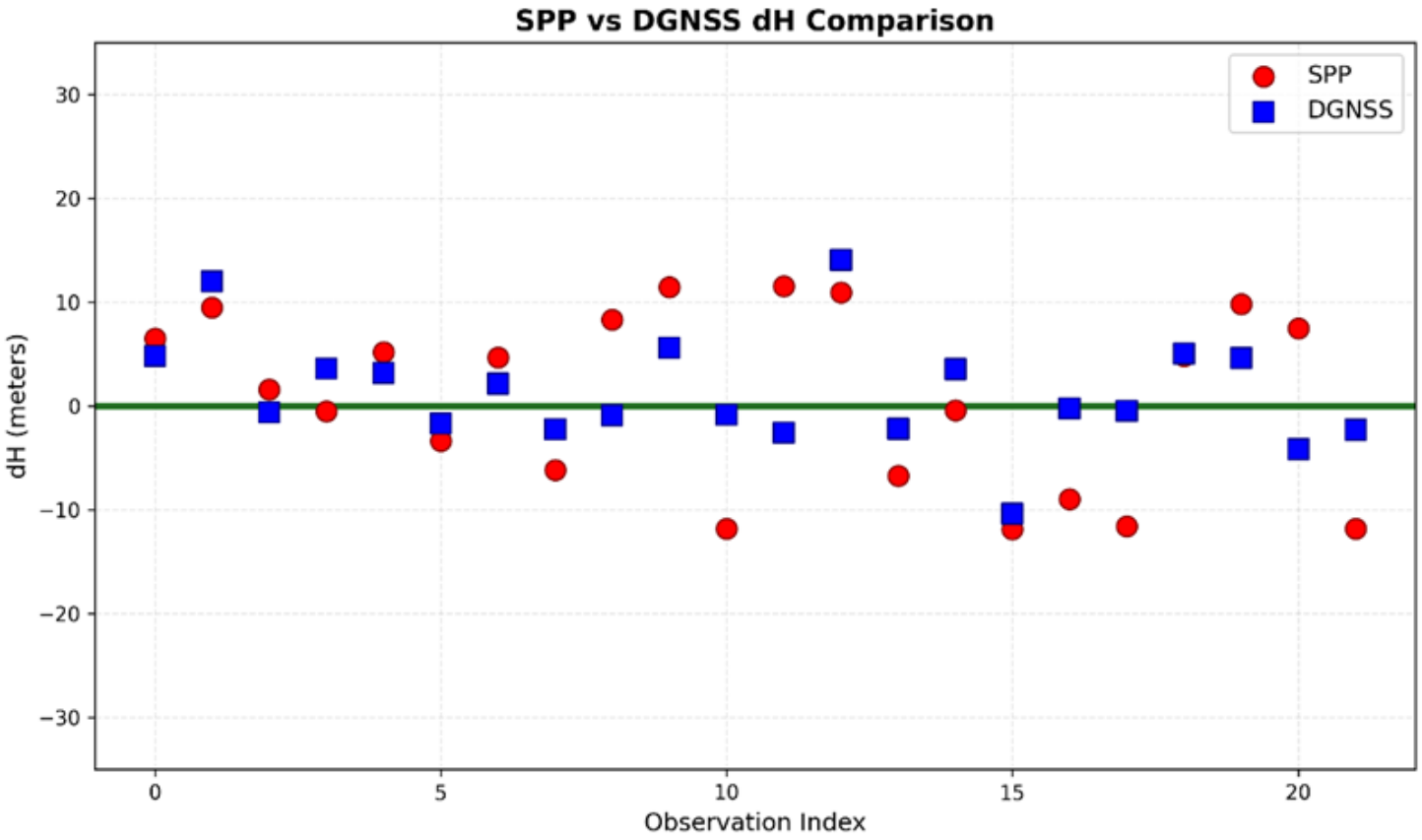
Overall accuracy summary

	SPP	**DGNSS**	Error reduction (improvement)
Horiz. MAE	7.39m	**5.26m**	**28.8%**
Vertical MAE	6.90m	**4.40m**	**36.3%**
Horiz. SDAE	4.58m	**3.13m**	**31.7%**
Vertical SDAE	3.80m	3.71m	2.3%
More accurate?	5 points	**17 points**	

Horizontal Error Distribution



Vertical Error Distribution



Discussion

1. Why DGNSS Improves
2. Why vertical improves more (36.3% vs 28.8%)
3. Why DGNSS sometimes underperforms SPP (5 station, 23%)
4. Limitations
5. Recommendations

Discussion - Why DGNSS improves?

- Atmospheric delays are spatially correlated (Tobler's First Law)

- SatRef base stations are close to receiver (city scale) -> assumptions valid

- Pseudorange corrections remove large iono/tropo biases

- Spatial cancellation more effective than SPP models

Discussion - Why vertical improves more?

- (36.3% vs 28.8%)
- Atmospheric delays affect vertical component more heavily
- Low-elevation satellites especially sensitive to troposphere
- Differential correction removes bias that models only partially mitigate

Discussion - Why DGNSS sometimes fails?

- SPP outperformed DGNSS in 5 stations (23%)
- Multipath is locally specific -> not cancelled by PRC
- Receiver noise is uncorrelated between base and rover
  - random offset
- District based (Sham Shui Po, Kowloon City) dense buildings -> likely multipath environment

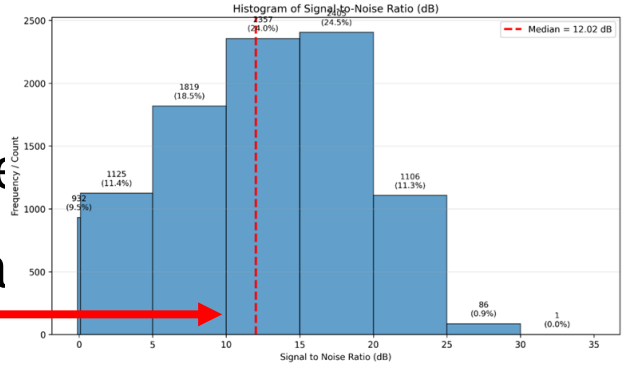
Limitations

- **Hardware**:

  - Single-frequency receiver (residual iono de

  - Low-cost antenna (high noise, SNR media

  - High Android clock drift



- **Algorithmic**:

  - GPS-only constellation (limited satellites in urban canyons)

  - Constant meteorology parameters (systematic SPP error)

- **Operational**

Recommendations (1/2)

- Priority 1: Multi-constellation (GPS + GLONASS + BeiDou)
  - Increase satellites available in urban areas
  - Impact: ~2–3 m accuracy improvement
  - SatRef already supports GLONASS corrections
- Priority 2: Dual-frequency device (L1 + L5)
  - Enable ionosphere-free pseudorange combination
  - Impact: correct residual ionospheric delay

Recommendations (2/2)

- Priority 3: Real-time meteorological data (HKO API)

  - Replace constant parameters in Saastamoinen model

- Priority 4: Measure actual antenna height on site

  - Avoid systematic 1 m offset assumption

Conclusion

- All Three Objectives Achieved

✓ Developed: Full GNSS pipeline (subframe decode, RTCM parse, WLS, HK1980 transform) on Android; code reproducible (GitHub)

✓ Evaluated: DGNSS achieved 28.8% horizontal, 36.3% vertical accuracy improvement over SPP; 77% of stations improved

✓ Recommended: Multi-constellation, dual-frequency, real-time meteorology, antenna height measurement

-> SatRef DGNSS corrections enable practical, metre-level positioning on consumer smartphones in Hong Kong

My GitHub Repository

github.com/Waterdragen/FinalYear

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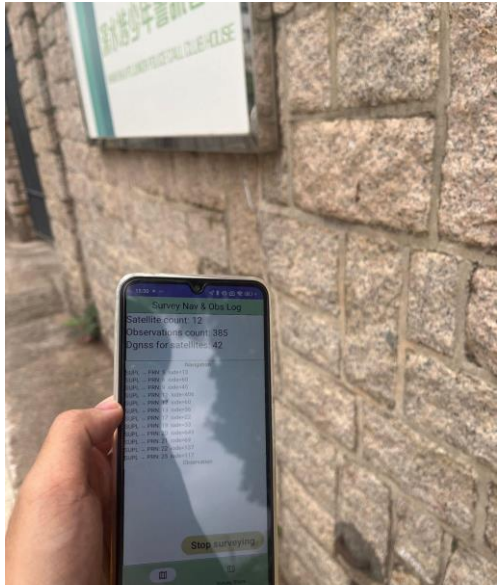
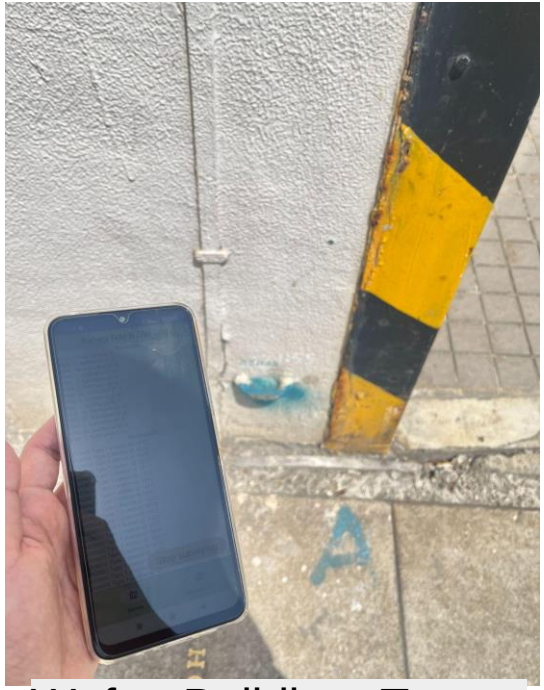
Thank you!

Any questions?



Sham Shui Po District Junior Police Call Clubhouse



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